

TECHBIZ SECURITY

CYBER SECURITY INTERNSHIP PROGRAM

MALWARE TYPES & COMMON ATTACKS

Internship Task Report — Week 3, Task 7 (SOC)

Submitted By	MANOJ
Role	Cyber Security Intern
Internship Organisation	TechBiz Security
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1. Introduction

This report documents the completion of Week 3, Task 7 of the SOC track within the TechBiz Security Cyber Security Internship. The task focuses on building foundational knowledge of common malware types and common cyber attacks, both of which form the basis for effective security monitoring, alert triage, and incident detection in a Security Operations Centre (SOC) environment.

Understanding malware is essential for a SOC analyst because malicious software is one of the primary mechanisms attackers use to gain access, maintain persistence, exfiltrate data, or disrupt operations. Similarly, familiarity with common cyber attacks such as phishing, denial-of-service, brute force, and man-in-the-middle attacks allows an analyst to recognise attack patterns as they unfold across logs, alerts, and network telemetry, rather than only after impact has occurred.

Malware and cyber attacks can compromise the confidentiality, integrity, and availability (CIA triad) of systems and data: confidentiality through unauthorised access or data theft, integrity through unauthorised modification of files or systems, and availability through disruption or denial of services. Recognising which part of the CIA triad is affected helps a SOC analyst prioritise and respond to incidents appropriately.

Security logs, indicators of compromise (IOCs), alerts, and unusual system or user behaviour are the primary evidence sources used to detect these threats. A single indicator is rarely sufficient on its own; SOC analysts typically correlate multiple signals across different log sources to confirm whether an event represents a genuine security incident.

2. Objectives

- Understand common malware types and their characteristics.
- Understand common cyber attacks used against organisations.
- Learn how different malware types operate and propagate.
- Understand how phishing attacks are constructed and delivered.
- Understand Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks.
- Understand brute-force and credential-based attacks.
- Understand Man-in-the-Middle (MITM) attacks.
- Understand how ransomware spreads and its operational impact.
- Identify basic detection signs and indicators associated with each attack type.
- Complete the required TryHackMe learning activities for this task.

3. Malware Types

The table below summarises the six malware types covered in this task, including how each operates, its typical impact, and the indicators a SOC analyst would look for during detection.

Malware Type	Definition	How It Works	Typical Impact	Detection Indicators
Virus	Malicious code that attaches itself to a legitimate file or program.	Requires execution or user interaction to activate and spread to other files.	File corruption, unauthorised code execution, system instability.	Unexpected file modifications; suspicious executable activity; antivirus/EDR alerts; unusual program execution.
Worm	Self-replicating malware capable of spreading independently.	Propagates across systems or networks without requiring the same level of user interaction as a virus.	Rapid host-to-host spread; network congestion; widespread compromise.	Suspicious network scanning; large numbers of connections; repeated suspicious processes across multiple systems.

Malware Type	Definition	How It Works	Typical Impact	Detection Indicators
Trojan	Malware disguised as legitimate or useful software.	Tricks the user into executing it, after which it performs malicious actions in the background.	Unauthorised access, data theft, deployment of further malicious payloads.	Malicious executable files; unauthorized processes; suspicious child processes; unexpected outbound connections; EDR/antivirus alerts.
Ransomware	Malware that encrypts or blocks access to data and demands payment.	Commonly spreads via phishing emails, malicious attachments, stolen credentials, vulnerable services, compromised systems, and lateral movement.	Data encryption; loss of system/data availability; business disruption; operational downtime; potential data theft; financial and reputational damage.	Large numbers of files being modified or encrypted; suspicious encryption processes; ransom notes; unusual file extensions; EDR alerts; abnormal file activity.
Spyware	Malware that secretly monitors users or systems.	Collects information such as activity, keystrokes, or credentials without authorisation.	Loss of confidentiality; credential exposure; unauthorised data collection.	Suspicious background processes; unexpected outbound network traffic; unknown connections to external systems.
Rootkit	Malware designed to hide malicious activity and maintain access.	Operates with privileged access to conceal files, processes, or its own presence from security tools.	Persistent unauthorised access; concealment of other malicious activity.	Hidden processes/files; system-level modifications; suspicious kernel/system activity; inconsistencies between security tools and system information.

4. Common Cyber Attacks

The following table outlines the common cyber attacks covered in this task, their mechanisms, typical detection signs, and their relevance to SOC monitoring.

Attack	Description	Mechanism	Detection Sign / Indicator	SOC Relevance
Phishing	A social-engineering attack using deceptive emails, messages, websites, or attachments.	Aims to achieve credential theft, malware delivery, or unauthorised access by tricking the recipient.	Suspicious sender address; look-alike domains; malicious URLs; unexpected attachments; urgent or suspicious messages; unusual authentication activity after credential theft.	Email security alerts; suspicious URLs; authentication anomalies; endpoint alerts; repeated similar phishing messages.
Denial of Service (DoS)	An attack that attempts to make a service unavailable.	Overwhelms or exhausts a system's resources through excessive requests, causing service degradation.	Sudden traffic spikes; high CPU/memory utilisation; large request volumes; service response degradation; connection exhaustion.	Indicates availability impact requiring immediate triage and possible mitigation via firewall or rate-limiting.
Distributed Denial of Service (DDoS)	A DoS attack carried out from multiple sources rather than one.	Uses multiple compromised or distributed systems to	Large traffic volume; many source IP addresses; sudden	Requires correlation across many sources;

Attack	Description	Mechanism	Detection Sign / Indicator	SOC Relevance
		generate traffic toward the target simultaneously.	request spikes; abnormal geographic/source distribution; network or server resource exhaustion.	often needs upstream or provider-level mitigation.
Man-in-the-Middle (MITM)	An attack where the attacker positions themselves between two communicating parties.	Intercepts or manipulates communication, potentially exposing credentials or data in transit.	TLS/certificate warnings; unexpected proxy behaviour; ARP anomalies; routing anomalies; unexpected network interception behaviour.	Highlights the importance of encrypted, authenticated communication channels.
Brute Force	Repeated attempts to guess passwords or authentication credentials.	Systematically tries credentials until valid access is obtained; may target one account or many.	Numerous failed login attempts; repeated authentication failures; one account targeted from multiple sources; one source targeting multiple accounts; successful login following many failures.	A key trigger for authentication log monitoring and account lockout policies.

5. Attack Detection from a SOC Perspective

SOC analysts identify malware and attacks primarily through continuous security monitoring across multiple log and telemetry sources, including authentication logs, Windows Event Logs, Linux logs, firewall logs, web server logs, IDS/IPS alerts, EDR alerts, antivirus alerts, network traffic, and email security alerts.

A single alert or log entry is rarely conclusive on its own. Effective detection depends on correlating multiple indicators across different sources to build confidence that an event represents a genuine incident rather than benign activity. The following examples illustrate this correlation approach:

- Phishing → suspicious email combined with a malicious URL and an unusual login event.
- Brute force → repeated failed logins followed by an eventual successful authentication.
- DoS/DDoS → an abnormal traffic spike combined with service degradation.
- Malware → a suspicious process combined with file changes and an outbound connection.
- Ransomware → mass file modifications combined with encryption behaviour and a ransom note.
- MITM → certificate, ARP, or network anomalies observed together.

6. Malware vs. Common Attacks

It is important to distinguish between related but distinct security terms used throughout this report:

- Malware – malicious software (such as a virus, worm, Trojan, ransomware, spyware, or rootkit) designed to damage, disrupt, or gain unauthorised access to a system.
- Cyber attack – a broader term describing any deliberate attempt to compromise a system, network, or data, which may or may not involve malware.
- Social engineering – a technique that manipulates human behaviour (for example, phishing) to obtain access or information, rather than exploiting a technical flaw directly.
- Network attack – an attack that targets network infrastructure or traffic, such as DoS, DDoS, or MITM.
- Credential attack – an attack that targets authentication mechanisms directly, such as brute force or password/hash cracking.

In summary, malware refers to the malicious software itself, while a cyber attack describes the method or technique used to compromise systems, users, credentials, or services — malware is often one of the tools used to carry out a wider attack.

7. TryHackMe Practical Exercises

To reinforce theoretical understanding, the following mandatory TryHackMe learning rooms were completed as part of this task. All exercises were performed within TryHackMe's authorised, isolated training environment, and no activity was conducted against real-world or production systems.

7.1 Common Attacks

Platform	TryHackMe
Status	Completed

Purpose of this room:

- Learn common cyber attacks.
- Understand social engineering and phishing.
- Understand malware and ransomware.
- Learn password and authentication attacks.
- Understand MITM concepts.
- Perform the room's interactive exercises.

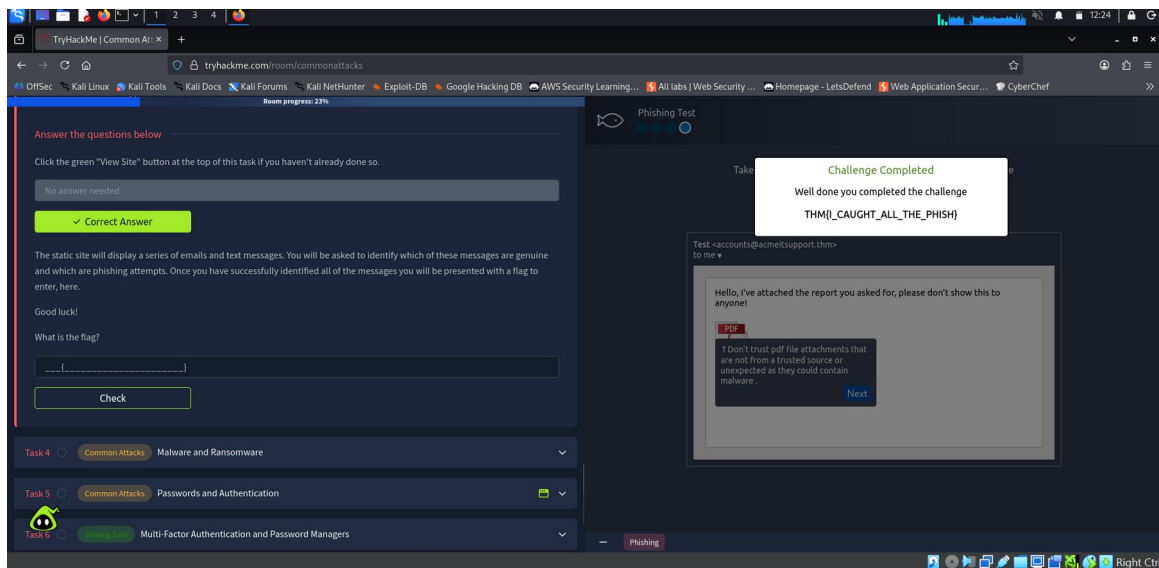


Figure 1: TryHackMe Common Attacks – Phishing challenge completed.

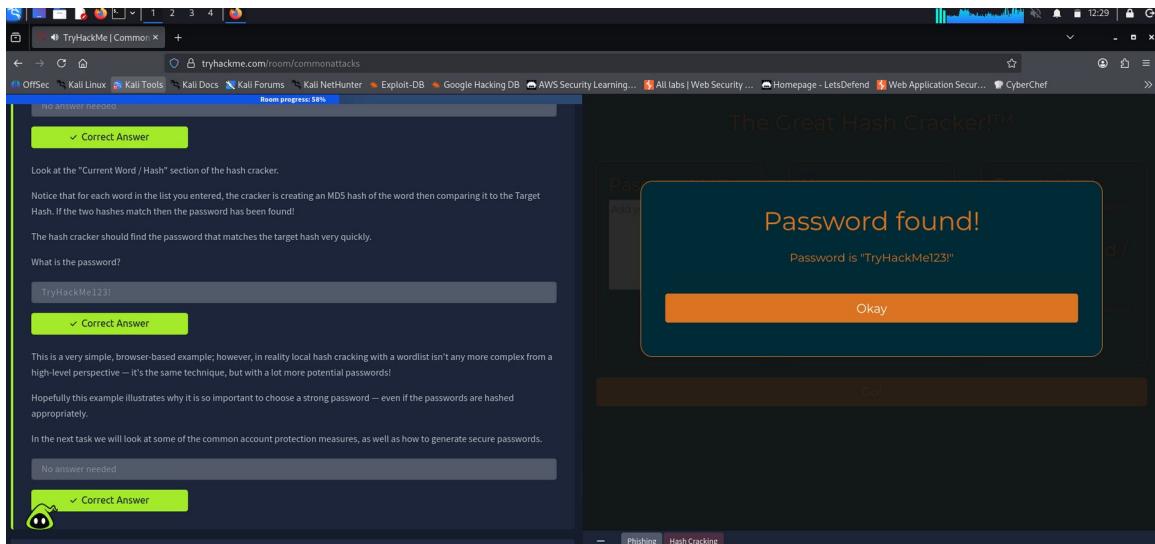


Figure 2: TryHackMe Common Attacks – Password/hash-cracking exercise completed.

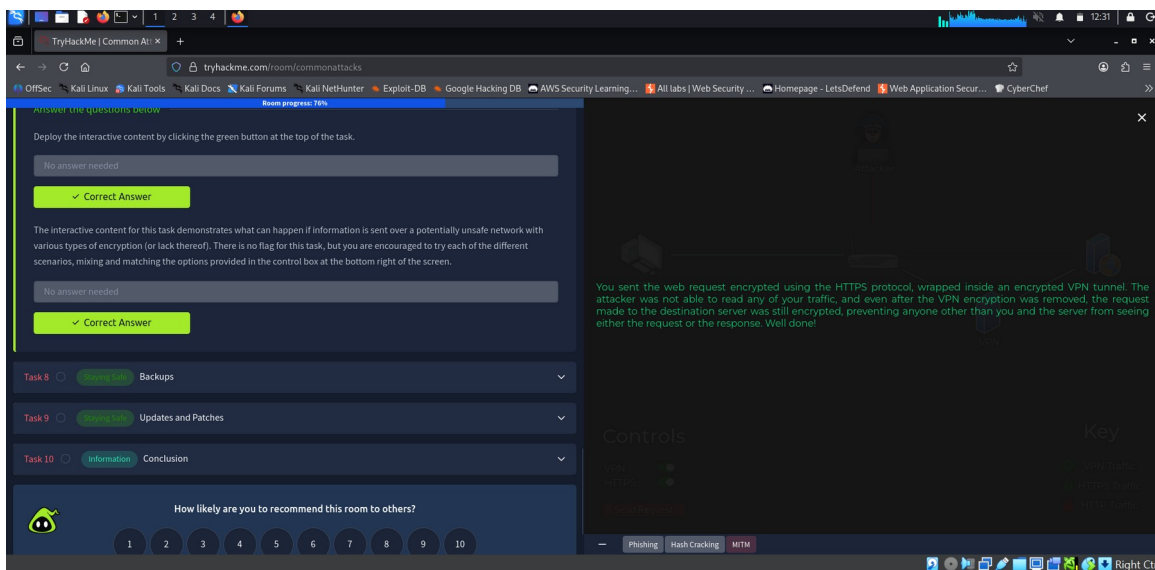


Figure 3: TryHackMe Common Attacks – MITM practical exercise demonstrating protected communication.

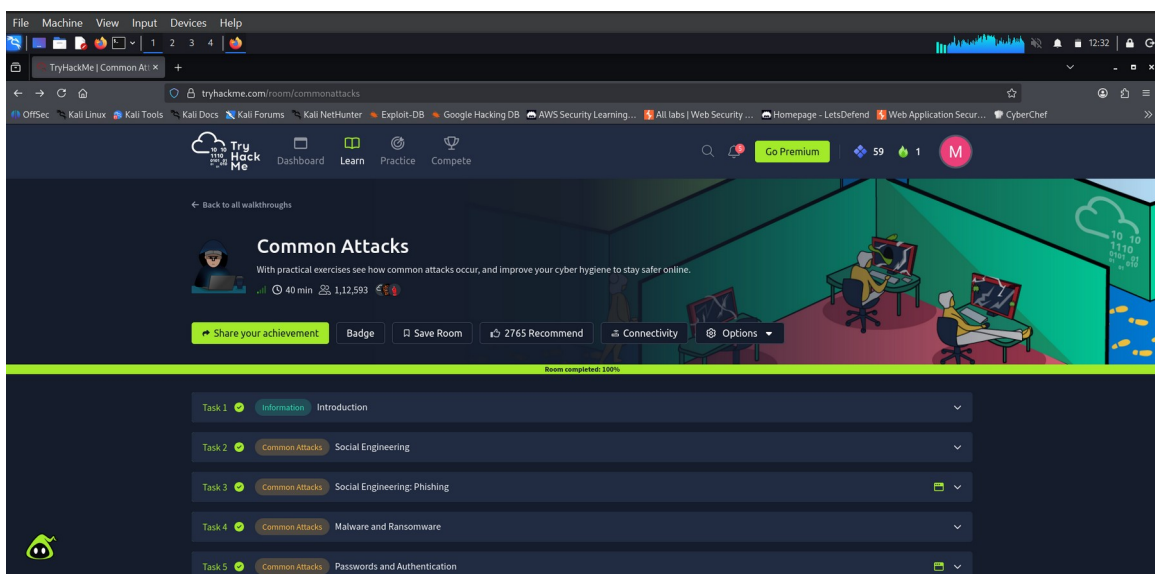


Figure 4: TryHackMe Common Attacks – Room completion evidence showing 100% completion.

7.2 Vulnerabilities 101

Platform	TryHackMe
Status	Completed

Purpose of this room:

- Understand vulnerabilities.
- Learn vulnerability concepts.
- Understand vulnerability scoring.
- Learn about vulnerability databases.
- Practice vulnerability research concepts.

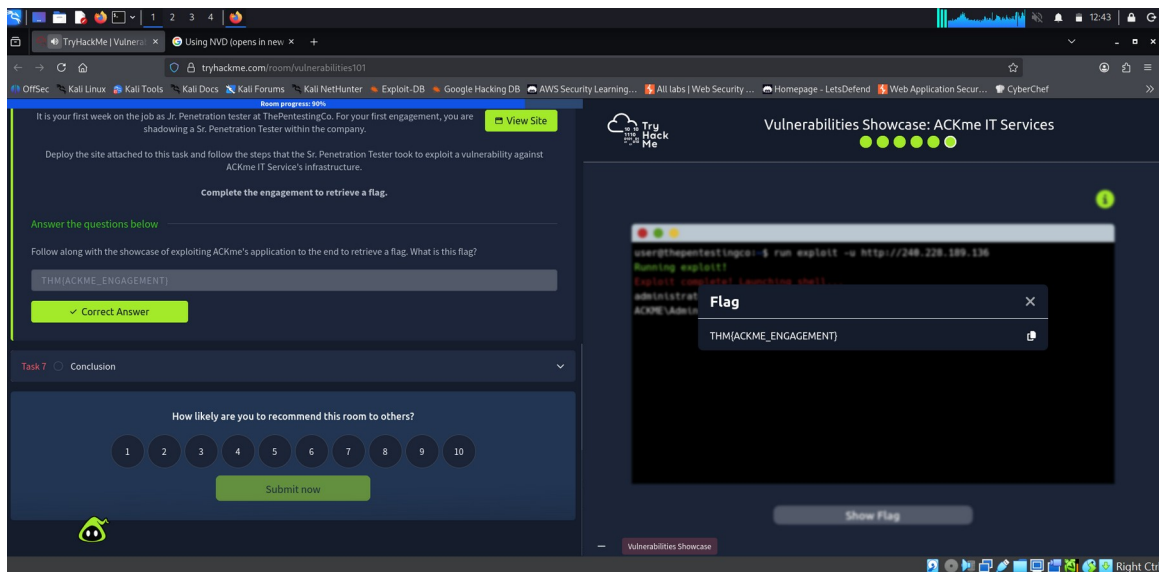


Figure 5: TryHackMe Vulnerabilities 101 – Practical vulnerability exploitation exercise with flag retrieval.

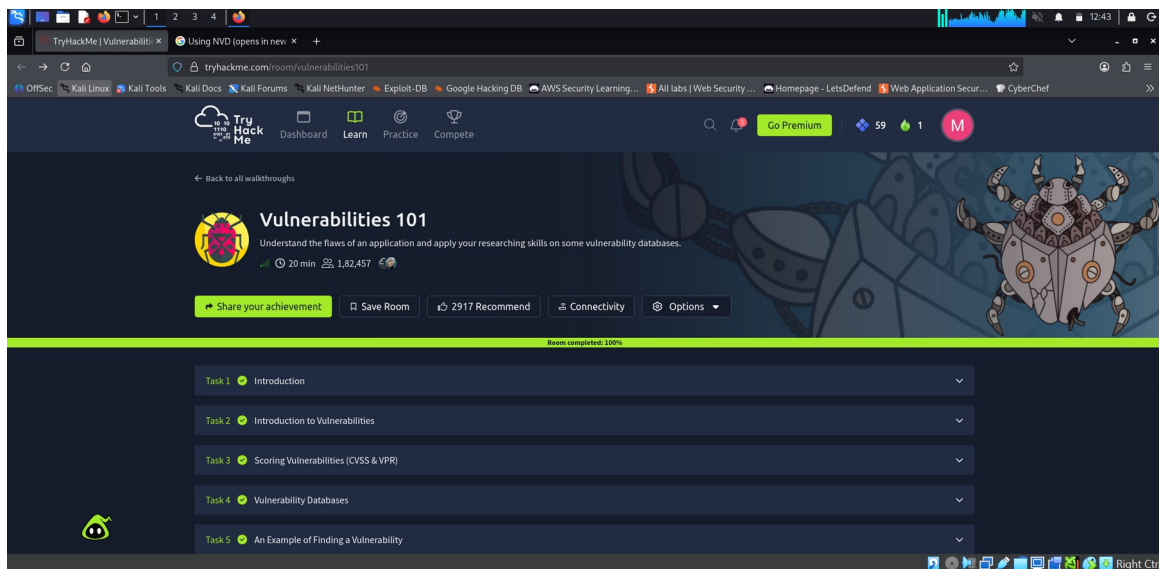


Figure 6: TryHackMe Vulnerabilities 101 – Room completion evidence showing 100% completion.

8. Practical Learning Outcomes

The TryHackMe exercises completed for this task provided practical reinforcement of the theoretical concepts covered above. Key outcomes include:

- Recognition of common attack patterns, including phishing and credential attacks.

- Understanding of phishing indicators, such as suspicious senders, attachments, and messaging.
- Understanding of password/hash-cracking concepts and the importance of strong passwords.
- Understanding of MITM risks and the value of secure, encrypted communication channels.
- Understanding of vulnerabilities and vulnerability research using standard databases and scoring systems.
- Importance of identifying indicators during security monitoring.
- Importance of correlating attack behaviour with logs and alerts rather than relying on isolated events.

It should be noted that these exercises were performed exclusively within TryHackMe's authorised training environment. No exploitation of real-world or production systems was performed or claimed as part of this task.

9. SOC Detection Cheat Sheet

The table below provides a concise reference summarising the primary detection indicator and relevant log/telemetry source for each threat covered in this report.

Threat	Primary Indicator	Possible Log/Telemetry
Virus	Unexpected executable/file changes	EDR / Antivirus / File logs
Worm	Rapid network propagation/scanning	Network / Firewall / EDR logs
Trojan	Suspicious executable and outbound connection	EDR / Network logs
Ransomware	Mass file encryption/modification	EDR / File / Windows logs
Spyware	Suspicious monitoring/data transmission	EDR / Network logs
Rootkit	Hidden/privileged activity	Endpoint / System security logs
Phishing	Malicious email/URL	Email / Proxy / Authentication logs
DoS/DDoS	Traffic spike/resource exhaustion	Firewall / Network / Server logs
MITM	Certificate/ARP/routing anomalies	Network / Endpoint logs
Brute Force	Repeated authentication failures	Authentication logs

10. Security Recommendations

- Use strong and unique passwords for all accounts.
- Enable multi-factor authentication (MFA) wherever possible.
- Implement email security and phishing protection controls.
- Keep operating systems and applications patched and up to date.
- Use antivirus/EDR solutions on endpoints.
- Monitor authentication logs for anomalous login activity.
- Implement continuous network monitoring.
- Maintain secure, regularly tested backups.
- Apply the principle of least privilege to user and system accounts.
- Segment critical systems from general network access.
- Monitor for abnormal file activity, particularly mass modifications.
- Conduct regular security awareness training for staff.
- Maintain and periodically test incident response procedures.

11. Conclusion

This task provided a structured foundation in identifying and understanding common malware types and cyber attacks relevant to SOC operations. Understanding how malware such as viruses, worms, Trojans, ransomware, spyware, and rootkits operate is essential for recognising the mechanisms attackers use to compromise systems and data.

Equally, understanding common attacks such as phishing, DoS/DDoS, MITM, and brute force enables a SOC analyst to identify malicious activity as it develops, rather than only after damage has occurred. The ability to recognise detection indicators and correlate them across multiple log sources is central to the role of a SOC analyst in monitoring, triaging, and responding to security incidents.

The TryHackMe practical exercises reinforced this theoretical knowledge through hands-on, authorised training scenarios covering phishing recognition, password/hash cracking, MITM communication risks, and vulnerability research. Completing this task has strengthened both conceptual understanding and practical awareness of common threats, and highlights the importance of continuous learning in the cybersecurity field as attack techniques and defensive practices continue to evolve.

12. Evidence / Screenshot Index

No.	Screenshot Filename	Description
1	01_common_attacks_phishing_challenge_completed.png	TryHackMe Common Attacks – Phishing challenge completed.
2	02_common_attacks_password_hash_cracking_completed.png	TryHackMe Common Attacks – Password/hash-cracking exercise completed.
3	03_common_attacks_mitm_practical_completed.png	TryHackMe Common Attacks – MITM practical exercise demonstrating protected communication.
4	04_tryhackme_common_attacks_room_completed.png	TryHackMe Common Attacks – Room completion evidence showing 100% completion.
5	05_vulnerabilities_101_flag_retrieved.png	TryHackMe Vulnerabilities 101 – Practical vulnerability exploitation exercise with flag retrieval.
6	06_tryhackme_vulnerabilities_101_room_completed.png	TryHackMe Vulnerabilities 101 – Room completion evidence showing 100% completion.